The strong-coupling gap of the Kogut–Susskind Hamiltonian is uniform in the volume

Statement T2 of the obligations map holds in Hamiltonian form: for every compact connected gauge group G and every lattice size, the Kogut–Susskind Hamiltonian at coupling $g \geq g_0(G)$ has a non-degenerate ground state and a gap

$$\Delta_{a,L} \;\geq\; \gamma\,\frac{g^2}{2}\,C_2(R_{\min})\,\frac{\hbar c}{a},$$

with $\gamma > 0$ independent of the lattice size, and the physical (gauge-invariant) gap is at least as large. The proof is a hypothesis check against Yarotsky's Theorem 1 on relatively bounded perturbations of classical lattice systems (Commun. Math. Phys. 261 (2006) 799, Sections 1 and Theorem 1 read at passage level): the electric term is a classical Hamiltonian in the Peter–Weyl partition with a product ground state and unit local gap after normalization, and the magnetic term is a bounded, translation-invariant, finite-range perturbation whose normalized size is $\beta(g) = 48N^2/[g^4(N^2-1)]$ for $SU(N)$, so the theorem's smallness condition $\beta \leq \beta_*(3, \Lambda_0)$ is the strong-coupling condition $g^4 \geq 48N^2/[(N^2-1)\beta_*]$. The same theorem gives the infinite-volume ground state and exponential clustering at these couplings. The constants β_* and γ are existential in Yarotsky's statement, so g_0 is a well-defined but not numerically explicit threshold. The result does not distinguish abelian from non-abelian groups; that distinction is T2′, and the rest of the problem is the descent of g_0 to zero along the scaling curve. No claim is promoted.

1. The Hamiltonian in Yarotsky's normalization

Notation as in the obligations map §1: periodic cubic lattice $\Lambda = (\mathbb{Z}/N_s\mathbb{Z})^3$, links $\mathcal{E}$, plaquettes $\mathcal{P}$, $\mathcal{H} = L^2(G^{\mathcal{E}})$, and

$$H = \frac{\hbar c}{a}\Big[\frac{g^2}{2}\sum_{\ell}(-\Delta_\ell) + \frac{2}{g^2}\sum_{p}\big(N - \operatorname{Re}\operatorname{tr} U_p\big)\Big].$$

Let $C_2 \equiv C_2(R_{\min})$ be the smallest nonzero Casimir eigenvalue of $-\Delta$ on $L^2(G)$; for $SU(N)$ with $\operatorname{tr} T^a T^b = \frac{1}{2}\delta^{ab}$, $C_2 = (N^2-1)/(2N)$. Divide by the unit $\varepsilon_0 = (g^2/2)C_2\,\hbar c/a$:

$$\frac{H}{\varepsilon_0} = \sum_{x\in\Lambda} h_x + \sum_{x\in\Lambda}\phi_x, \qquad h_x = \frac{1}{C_2}\sum_{i=1}^{3}(-\Delta_{(x,x+e_i)}), \qquad \phi_x = \frac{4}{g^4 C_2}\sum_{i<j}\big(N - \operatorname{Re}\operatorname{tr} U_{p(x,i,j)}\big),$$

where $p(x,i,j)$ is the plaquette with corner x spanned by e_i, e_j. Every link belongs to exactly one h_x and every plaquette to exactly one ϕ_x, so the two sums reproduce H/ε_0.

2. Hypothesis check

Yarotsky's setting (Section 1 of the paper): a lattice $\mathbb{Z}^\nu$, site spaces $\mathcal{H}_x$ possibly infinite-dimensional with a preferred vector Ω_x, translation-invariant interactions of finite range Λ_0, a classical part $H_0 = \sum_x h_x$ where h_x is a function of a product partition of unity containing the projection onto Ω_x and satisfies $h_x\Omega = 0$, $h_x \geq 1$ on the orthogonal complement of the product vector, and a perturbation $\sum_x \phi_x$ with each ϕ_x a symmetric form on $\mathcal{H}_{\Lambda_0+x}$ obeying $|\phi_x(v,v)| \leq \alpha\|h_x^{1/2}v\|^2 + \beta\|v\|^2$ with $\alpha < 1$. Theorem 1 there: there are $\alpha_*, \beta_* > 0$ depending only on ν and Λ_0 such that if $\alpha \leq \alpha_*$ and $\beta \leq \beta_*$, then for periodic cubic volumes H_Λ has a non-degenerate ground state with a gap $\gamma > 0$ independent of Λ, the ground states have a weak* thermodynamic limit, and that limit has exponentially decaying correlations.

1. *Lattice and site spaces.* $\nu = 3$; $\mathcal{H}_x = L^2(G)^{\otimes 3}$, the three links leaving x; Ω_x the constant function. Then $\mathcal{H}_\Lambda = \bigotimes_x \mathcal{H}_x = L^2(G^{\mathcal{E}})$ and $\Omega_{\Lambda,0}$ is the constant function on $G^{\mathcal{E}}$.

2. *Partition and classicality.* On each link, Peter–Weyl gives the orthogonal partition of unity by isotypic projections P_R, R ranging over irreducible representations, with P_{triv} the projection onto constants. $-\Delta_\ell = \sum_R C_2(R) P_R$ is a function of this partition, so h_x is a function of the product partition on $\mathcal{H}_x$, and h_x is self-adjoint and unbounded, as allowed.
3. *Local ground state and gap.* $h_x \Omega_x = 0$. On the orthogonal complement of Ω_x in $\mathcal{H}_x$ at least one of the three links carries a nontrivial representation, so $h_x \geq C_2(R_{\min})/C_2 = 1$ there: condition (1) of the paper holds with the range $\Lambda_0 = \{0\}$ for h_x, which may be enlarged to the common range below.
4. *Perturbation.* ϕ_x is multiplication by a continuous function on the twelve links of the three plaquettes at x, whose sites lie in $\Lambda_0 + x$ with $\Lambda_0 = \{0,1\}^3$; it is symmetric, bounded, and $0 \leq N - \mathrm{Re}\,\mathrm{tr}\, U_p \leq 2N$ gives

$$|\phi_x(v,v)| \leq \beta \|v\|^2, \qquad \beta = \frac{4}{g^4 C_2} \cdot 3 \cdot 2N = \frac{24N}{g^4 C_2} = \frac{48N^2}{g^4(N^2-1)} \quad (SU(N)),$$

   so condition (2) holds with $\alpha = 0$ and this β.
5. *Translation invariance and periodicity.* h_x and ϕ_x are translates of h_0 and ϕ_0, and the lattice is a periodic cube, the case in which the paper states Theorem 1.

Theorem (T2, Hamiltonian form). Let $\beta_* = \beta_*(3, \{0,1\}^3)$ and γ be the constants of Yarotsky's Theorem 1 for this range. For every g with $48N^2/[g^4(N^2-1)] \leq \beta_*$, that is $g \geq g_0(N) = \left(48N^2/[(N^2-1)\beta_*]\right)^{1/4}$, and every $N_s \geq 2$: H has a non-degenerate ground state, and

$$\Delta_{a,L} \;\geq\; \gamma\, \varepsilon_0 = \gamma\, \frac{g^2}{2}\, \frac{N^2-1}{2N}\, \frac{\hbar c}{a}.$$

The physical gap obeys $\Delta_{a,L}^{\mathrm{phys}} \geq \Delta_{a,L}$ by Theorem 1(3) of the obligations map. The ground states have a thermodynamic limit ω with exponentially decaying correlations of local observables.

Proof. Items 1–5 are the hypotheses of Yarotsky's Theorem 1 with $\alpha = 0 \leq \alpha_*$ and $\beta \leq \beta_*$; its conclusions 1–3 are the statements, after multiplying the normalized gap γ by ε_0. The inheritance by the physical sector is the positivity argument already written for T1. □

For a general compact connected G the same holds with $C_2 = C_2(R_{\min})$ and the bound $N - \mathrm{Re}\,\mathrm{tr}\, U_p \leq 2\,\dim$ in place of $2N$; the threshold is $g_0^4 = 24\,\dim/(C_2 \beta_*)$.

3. The abelian comparison and a finite-volume bound

For compact $U(1)$ take magnetic term $(2/g^2)\sum_p(1-\cos\theta_p)$ and electric term $(g^2/2)\sum_\ell n_\ell^2$. On a periodic cubic lattice with $N_s \geq 4$, Gauss's law forces a nonzero charge network to contain a cycle, whose shortest length is four. A plaquette loop realizes it, so the physical electric gap is exactly $2g^2\hbar c/a$. With $P = 3N_s^3$, subtracting the scalar magnetic constant leaves an operator of norm at most $2P/g^2$ in units $\hbar c/a$. Min–max bounds the two lowest eigenvalue shifts separately and therefore

$$\left(2g^2 - \frac{4P}{g^2}\right)\frac{\hbar c}{a} \leq \Delta_{a,L}^{\mathrm{phys}} \leq \left(2g^2 + \frac{4P}{g^2}\right)\frac{\hbar c}{a}, \qquad L = N_s a.$$

At fixed N_s, the ratio to $g^2\hbar c/a$ tends to 2 as $g \to \infty$; the lower bound gives no volume-uniform threshold. The corresponding $SU(3)$ finite-volume calculation and the incomplete continuous-time polymer count are in the continuous-time note.

The existential Yarotsky theorem applies to both groups. Their weak sides differ: for $SU(3)$ the perturbative slope is $dg^2/d\log a = 2b_0 g^4 + O(g^6)$, $b_0 = 11/(16\pi^2)$, while the continuum pure photon has no perturbative self-interaction. Guth (Phys. Rev. D 21 (1980) 2291) and Fröhlich–Spencer (Commun. Math. Phys. 83 (1982) 411) are metadata-level sources for a weak-coupling massless phase of compact lattice $U(1)$. Perturbative running is not an exact compact-lattice blocking identity and does not establish a single gapped interval, controlled arrival at the target box or continuum reconstruction. The former uniform 4λ rate and numerical box location are withdrawn; the target box is existential.

4. What the theorem does and does not give

- *Uniformity.* γ depends only on $\nu = 3$ and $\Lambda_0 = \{0,1\}^3$, so the bound holds for every lattice size at once. The extensive Kato bound of the obligations map §3 is replaced by a local one, which is exactly what the cluster structure of the perturbation provides.
- *Large N.* $\beta \to 48/g^4$ as $N \to \infty$, so $g_0(N)$ is bounded in N: the strong-coupling gap is uniform in the rank at fixed bare g in this convention. In the 't Hooft normalization $\lambda = g^2 N$ the threshold reads $\lambda \geq g_0^2 N$, growing with N.
- *Scale.* The gap is measured in $\hbar c/a$ and is proportional to g^2: a supplied cutoff scale, as the dimensional census of G07 §4 requires at fixed g. Nothing here survives $a \to 0$ at fixed g, and T3 asks for the behaviour along $g(a) \to 0$, outside the theorem's range.
- *Abelian case.* $U(1)$ satisfies every hypothesis with $C_2(R_{\min}) = 1$ (charge one) and dim $= 1$, so it too is gapped uniformly in volume at strong coupling. The theorem is blind to the property that distinguishes the groups; that property enters only in T2′, where the abelian gap closes (Guth; Fröhlich–Spencer) and the non-abelian one is conjectured to persist.
- *Explicitness.* β_* and γ exist by the theorem and are not given as numbers in its statement. A heuristic extraction of the time-discretized expansion gives $\beta \simeq e^{-465}$ and g_0^2 of order 10^{102} for $SU(3)$, but the required support-weight bound is unproved; the continuous-time expansion of the Kogut–Susskind note gave $g_0^2 = 388$ (79 in the adjacent-growth class) with $\gamma \to 4$; that threshold was withdrawn on 2026-10-02 after its overlap count was found incomplete, so no explicit KS number stands. The descent problem does not depend on that number, only on the structure of the expansion.
- *Relation to the small-volume end.* Yarotsky's expansion is in the electric basis, where the ground state at $g = \infty$ is the constant function. At weak coupling the ground state concentrates near $U_p = 1$ and the good variables are the constant modes of G07 §3.3, with gap $\delta_1 g^{2/3} \hbar c/L$ in the continuum small box. The two expansions use different bases and different small parameters (g^{-4} and $g^{2/3}$); T2′ is the statement that the gap does not close between them.

5. Consequence for STATE

T2 is closed in Hamiltonian form as a corollary of an established theorem, with the threshold $g_0(N) = (48N^2/[(N^2-1)\beta_*])^{1/4}$. The obligations table moves T2 to "established"; the open items are T2′, T3 and T4. The next theorem-sized target on the lattice route is a weak-coupling counterpart: a variational upper bound and, if possible, a lower bound on the Kogut–Susskind gap at small g on a lattice whose side is small against $\hbar c/\Lambda$, using the constant-mode ground state of C133 as the trial state, to connect the lattice gap $\Delta_{a,L}$ with the continuum small-volume corner and to locate the regime where neither expansion applies.